INCH-POUND

MIL-DTL-18714F (OS)
19 February 2012
SUPERSEDING
MIL-DTL-18714E
10 November 2011

DETAIL SPECIFICATION

PRIMER, ELECTRIC, MARK 45 MOD 1

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the minimum requirements for the Electric Primer, Mark 45 Mod 1, herein referred to as the "primer."

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-3464 Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

Comments, suggestions, or questions on this document should be addressed to Department of the Navy, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

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DEPARTMENT OF DEFENSE SPECIFICATIONS – continued

MIL-T-15119	Tubing, Round, Seamless Alloy Steel
MIL-P-223	Powder, Black
MIL-V-16399	Varnish, Moistureproof
MIL-L-296	Lacquer, Purple
MIL-I-695	Insulation, Electrical, Paper (Slot-Cell)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DOD Preferred Methods for Acceptance of Products
DOD-STD-2101	Classification of Characteristics

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE CODE 10001 AND 53711)

2434755	Primer, Electric, MARK 45 MOD 1
3208375	Charge, Propelling, 5-Inch, 54 Caliber, Full, MARK 67 MOD 3

(Application for copies of NAVSEA drawings should be addressed to: US Navy Major Caliber Propelling Charge AEA, Naval Surface Warfare Center Indian Head Division Picatinny Detachment, Attn: G2, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742/E1742M	Standard Practice for Radiographic Examination

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ASTM D5363	Standard Specification for Anaerobic Single-Component Adhesives (AN)
ASTM B633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM A108	Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A434	Standard Specification for Steel Bars, Alloy, Hot-Wrought or Cold-Finished, Quenched and Tempered
ASTM A193/A193M	Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications
ASTM D707	Standard Classification System and Basis for Specification for Cellulose Acetate Butyrate Molding and Extrusion Compounds
ASTM B152/B152M	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM B36/B36M	Standard Specification for Brass Plate, Sheet, Strip, And Rolled Bar
ASTM B16/B16M	Standard Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines
ASTM D4066	Standard Classification System for Nylon Injection and Extrusion Materials (PA)
ASTM D6779	Standard Classification System for Polyamide Molding and Extrusion Materials (PA)

(Copies of these documents are available online at http://www.astm.org or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

IPC J-STD-006	Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications
IPC J-STD-004	Requirements for Soldering Fluxes
IPC J-STD-001	Requirements for Soldered Electrical and Electronic Assemblies

(Copies of these documents are available online at http://www.ipc.org/default.aspx or from the Institute for Interconnecting and Packaging Electronic Circuits Customer Service, 3000 Lakeside Drive, 309S, Bannockburn, IL 60015.)

2.4 Order of precedence. Unless otherwise specified herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article and quality conformance. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3. Primer lots shall be formed in accordance with 4.4.1. All primer lots shall be subjected to quality conformance inspection in accordance with 4.4.

3.2 Characteristics. All primers shall be inspected according to Tables II through XXVII as specified herein. The characteristics of the primer assembly drawing (drawing 2434755), and associated components, shall be inspected for conformance in accordance with the quality requirements identified in the contract, which shall include the requirements in 4.3 and 4.4.

3.3 Radiographic (M101E). All primers shall be 100% radiographically inspected using the methods specified in 4.6.1. Primers shall not exhibit any discernible gaps as identified in Figure 1 (locations 1, 2, and 3). The presence, correct position, and correct orientation of the components of drawing 2434755 shall be verified (except surface primer, sealing compound, and varnish, find nos. 4, 5, and 8, respectively).

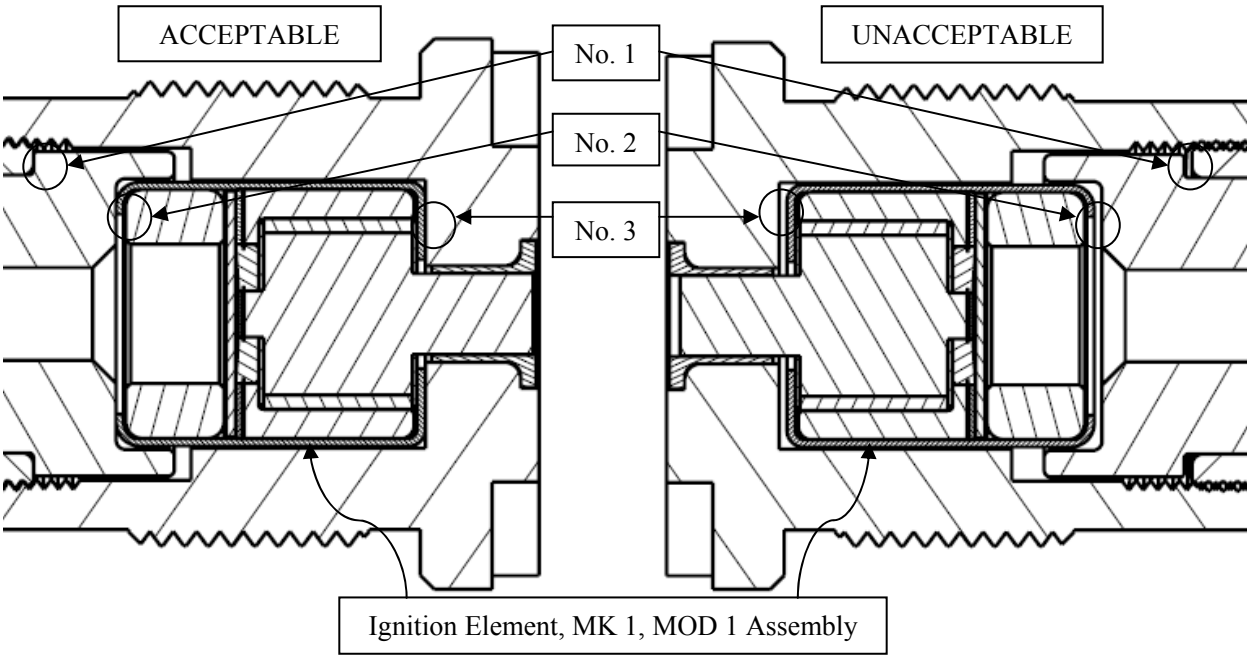


FIGURE 1. Gap locations

3.4 Black powder moisture content (M102). The moisture content of the black powder within the primer shall not exceed 0.70 percent by weight (see 4.6.2).

3.5 Detail strip. When required for first article inspection (see 4.3), and quality conformance inspection (see 4.4), the primer shall undergo detail strip inspection in accordance with 4.6.3 to verify conformance of 3.5a and 3.5b, below. Electrical resistance (see 3.6) and vibration (see 3.8) shall be performed prior to detail strip for the applicable primers.

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a. (C1) Each critical characteristic shall be as identified in Table XXVIII.

b. (M103) Each major characteristic shall be as specified in Table XXVIII.

3.6 Electrical resistance (M104E). All primers shall be 100% inspected using the methods specified in 4.6.4. The primer shall have an electrical resistance of 0.10 to 0.18 ohms.

3.7 Functional. The primer shall meet the following criteria (see 4.6.5):

a. (M105) Primer functioning time (see 4.6.5.3), shall be a maximum of 0.006 second;

b. (M106) The primer shall not misfire (see 6.7.1);

c. (M107) The primer shall not exhibit blowback (see 6.7.2);

d. (C2) Except for the blowout of the ignition element closing discs (drawing 2846969), the primer shall not exhibit a split stock, split tube, or detachment of metallic components.

3.8 Vibration. When required for first article inspection, the primer shall meet the following requirements (see 4.6.6):

a. (C3) The primer shall not function during vibration;

b. (C4) The stock and tube shall maintain material integrity and the primer shall maintain assembly integrity at the mating threads of the stock and tube during vibration;

c. (M108) The primer shall not otherwise physically breakdown during vibration;

d. The primer shall meet the requirements of 3.6 after vibration;

e. The primer shall meet the requirements of 3.7 after vibration.

3.9 Ballistics. When assembled into a 5"/54 propelling charge (drawing 3208375) for first article inspection, the primer shall provide ignition, which will meet the criteria of 3.7 b, c, and d, and the following (see 4.6.7):

a. (M109) The ejection time of the projectile from the gun barrel shall not be greater than 0.050 second after application of firing energy.

3.10 Workmanship. All parts shall be free of chips, dirt, grease, rust, damage, and all other foreign materials. There shall be no burrs or other imperfections of manufacture such as cracks, sharp edges, unblended radii, and surface defects which would affect the assembly functioning or handling of the product. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All markings shall be legible.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.2 Test and examination methods. Primers shall be selected at random for each test or examination. Identification of each primer's complete test and examination results shall be secured for additional examination until otherwise directed by the contracting activity (see 6.6).

4.3 First article inspection. The first article shall be a homogeneous (6.7.7) sample of 130 assembled primers and five-each of every component part and sub-assembly (less explosive components) fabricated, packaged, and packed in accordance with the requirements of this specification and the contract or purchase order using the methods and procedures to be used for production. The first article shall be submitted to the government facility designated in the contract or order for test and examination in accordance with the first article inspection plan of Table I (see 6.2 and 6.3). Prior to delivery, a first article shall be inspected and tested for all requirements specified herein.

4.3.1 Five-each inspection. 5 units of every part and non-explosive sub-assembly shall be selected from a population that is representative of parts to be used for standard primer production. Each of these parts and sub-assemblies shall be inspected for compliance with the requirements of their respective designs (drawings and specifications). These shall be the same inspections that are performed during quality conformance for parts and sub-components (see 4.4.3 and Tables II through XXVIII).

4.3.2 First article rejection. Failure of the first article sample to meet any of the requirements shall be cause for failing first article.

4.4 Quality conformance inspection.

4.4.1 Lot formation. Unless otherwise specified in the contract or purchase order (see 6.2), the production lots shall consist of not more than 20,000 homogeneous (see 6.7.7) primers plus those required for inspection (see 4.4.2). Each production lot shall contain only one Class 2 black powder lot and one Class 4 black powder lot (IAW MIL-P-223) and only one tubing (IAW MIL-DTL-15119) heat treat lot and one ignition element lot.

4.4.1.1 Ignition element lot. An ignition element lot shall consist of a homogenous production of MK 1 MOD 1 Ignition Elements (drawing 2434753). The following clarifications apply:

a. A change in the bridge-wire manufacturer's lot shall lead to a change in the ignition element lot;

b. For a single lot of ignition elements, the ignition charge (drawing 2434757) may be manufactured in small batches if the batches come from the same manufacturer using the same lots of ingredients and are manufactured with no break in production for longer than 20 working days.

4.4.2 Primer inspection. Unless otherwise specified in the contract or purchase order (see 6.2), 105 samples shall be selected at random from each production lot for test and examination at the government facility designated in the contract or order in accordance with the quality conformance inspection plan of Table I. Conformance inspection shall be performed in accordance with Tables II through XXVII as specified

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herein. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests. This sample is in addition to those primers required for contractor inspection and tests.

4.4.3 Part and sub-component inspection. Unless otherwise specified in the contract or purchase order (see 6.2), or on tables II through XXVIII, quality conformance of parts and sub-components shall be determined in accordance with Tables II through XXVIII and MIL-STD-1916. Unless otherwise specified, characteristics identified as CRITICAL shall be inspected 100%. Unless otherwise specified, inspection level IV shall be used for all characteristics identified as MAJOR and inspection level II for characteristics identified as MINOR.

4.5 Classification of characteristics. The characteristics verified by the tests and examination herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The symbol (E) designates a requirement for exceptional testing or inspection that is not covered by a standard sampling plan. The number following the classification symbol indicates the serial number of the characteristic. Requirements not annotated with a classification symbol are classified as minor. (Reference tables II through XXVIII).

4.5.1 Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless otherwise specified, inspection level IV shall be used for all characteristics defined as major and inspection level II for all minor characteristics. Critical characteristics shall be addressed in accordance with MIL-STD-1916. In addition to the sampling requirements herein, verification level VII shall be used to verify the 100% screening operation of critical characteristics.

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TABLE I. First article and quality conformance inspection plan

Test or Examination	Requirement Paragraph	Test Paragraph	First Article Inspection			Quality Conformance Inspection		
			Sample Size	Accept	Reject	Sample Size	Accept	Reject
Radiographic	3.3	4.6.1	130[1/]	0	1	105[1/]	0	1
Electrical resistance	3.6	4.6.4	130[1/]	0	1	105[1/]	0	1
Functional[3/] Functioning time Misfire Blowback Split stock, split tube, or parts detached	3.7 a. b. c. d.	4.6.5	30	 0 0 0 0	 1 1 1 1	25	 0 0 0 0	 1 1 1 1
Vibration[3/] Non-functional Integrity of stock, tube, assembly, etc.	3.8 a. b., c. d. e.	4.6.6	9	 0 0 0 0	 1 1 1 1	0	 N/A N/A N/A N/A	 N/A N/A N/A N/A
Ballistic ejection time[3/]	3.9	4.6.7	20	0	1			
Detail Strip[3/] Critical characteristics Major characteristics	3.5	4.6.3	56	 0 0	 1 1	56	 0 0	 1 1
Black powder moisture content	3.4	4.6.2	5[2/]	0	1	5[2/]	0	1
Parts and sub-assemblies	3.1	4.3.1 and 4.4.3	5 each	0 per item	1 per item	MIL-STD-1916 and Tables II through XXVIII		
Packaging	Section 6.4.1	4.6.8	1 pk	0	1	1 pk	0	1

[1/] Radiographic examination and resistance test shall be performed on all samples prior to any other test or examination.
[2/] Samples shall be taken from those for the detail strip examination.
[3/] Performed only by the government facility as designated in the contract or order.

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TABLE II. Conformance inspection by classification of characteristics for primer stock, drawing number 1275047

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	.750-32 UN-2B	Level IV	3.2	AIE
M102	1.00-20 UNEF-2A -C-	Level IV	3.2	AIE
-M103	Ø .7162 +.0034/-.0000	Level IV	3.2	AIE
M104	.254 +.000/-.010	Level IV	3.2	AIE
+M105	Ø 1.187 +.000/-.005	Level IV	3.2	AIE
M106	Note 3: Zinc plate IAW ASTM B633, SC 1, Type II; maximum thickness of 20 µm. Internal thread is not subject to minimum thickness requirement, but shall show evidence of coating.	Level IV	3.2	AIE & COC
M107	⊥ .002 C(M) -A-	Level IV	3.2	NSIG
M108	⌖[(Ø .002(M)][C(M)] -E-	Level IV	3.2	NSIG
-M109	.625 ± .020	Level IV	3.2	AIE
M110	.740 min depth full threads	Level IV	3.2	NSIG
M111	⌖[Ø .002(M)][C(M)[A] -F-	Level IV	3.2	NSIG
-M112	4 x R .185 +.004/-.000	Level IV	3.2	AIE
+M113	Ø .208 +.003/-.000 thru	Level IV	3.2	AIE
-M114	45° x .03	Level IV	3.2	AIE
-M115	.100	Level IV	3.2	AIE
M116	⌖[Ø .010(M)][E(M)]	Level IV	3.2	AIE
+M117	Ø .579 +.003/-.000	Level IV	3.2	AIE
M118	⌖[Ø .003(M)][D(M)]	Level IV	3.2	NSIG
M119	.11 max	Level IV	3.2	AIE
+M120	Ø .930 +.000/-.010 2 places	Level IV	3.2	AIE
M121	⌖[Ø .005(M)][C(M)][A]	Level IV	3.2	NSIG
M122	⌖[Ø .005(M)][F(M)] -D-	Level IV	3.2	NSIG
+M123	1.763 ± .010	Level IV	3.2	AIE

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TABLE II. Conformance inspection by classification of characteristics for primer stock, drawing number 1275047 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Major				
+M124	.502 +.000/-.004	Level IV	3.2	AIE
M125	R .010 max	Level IV	3.2	AIE
Minor				
201	Note 4: Material: steel IAW ASTM A108, grades 1213 or 1215, or material: heat treated alloy steel bar, cold finished, 1.250 inch diameter, per ASTM A434, grade 4140, class BC or per ASTM A193/A193M, grade B7, hardness Rockwell C 28-32	Level II	3.2	COC
202	Ø .320 +.003/-.000, counterbore-depth .042 ± .003	Level II	3.2	AIE
203	⌖ [Ø .001(M)][B(M)] (for Ø .320 +.003/-.000)	Level II	3.2	AIE
204	R .035 +.015/-.000	Level II	3.2	AIE
205	1.094	Level II	3.2	AIE
206	// .002 A	Level II	3.2	AIE
207	45° x .02	Level II	3.2	AIE
208	45° x .03	Level II	3.2	AIE
209	Start thread 45° angle	Level II	3.2	AIE
210	End thread 45° angle	Level II	3.2	AIE
211	⌖ [Ø .001(M)][F(M)] (for Ø .208 +.003/-.000)	Level II	3.2	AIE
212	Note 2: Surface texture, √ 125	Level II	3.2	AIE
213	Note 2: Remove all burrs	Level II	3.2 & 3.10	AIE
214	Note 2: Fillets, R .01 max	Level II	3.2	AIE
215	Note 2: Outside corners, 45° x .01 max	Level II	3.2	AIE
216	Note 2: Dimensions apply after plating.	Level II	3.2	AIE
217	.243 + .000/-.005	Level II	3.2	AIE

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TABLE II. Conformance inspection by classification of characteristics for primer stock, drawing number 1275047 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Minor				
218	Ø .579 -.000 (Lower Tol)	Level II	3.2	AIE
219	.625 +.020 (Upper Tol)	Level II	3.2	AIE
220	Ø .930 -.010 (Lower Tol)	Level II	3.2	AIE
221	45° + 2° x .03 + .01 (Upper Tol)	Level II	3.2	AIE
222	1.763 -.005 (Lower Tol)	Level II	3.2	AIE
223	.208 -.000 (Lower Tol)	Level II	3.2	AIE
224	Ø 1.187 -.005 (Lower Tol)	Level II	3.2	AIE
225	4 x R .185 +.004 (Upper Tol)	Level II	3.2	AIE
226	.100 + .005 (Upper Tol)	Level II	3.2	AIE
227	Ø .930 -.010 (Lower Tol)	Level II	3.2	AIE
228	Ø .7162 +.0034 (Upper Tol)	Level II	3.2	AIE
229	.502 -.004 (Lower Tol)	Level II	3.2	AIE

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TABLE III. Conformance inspection by classification of characteristics for primer stock insulator, drawing number 451563

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	.191± .002	Level IV	3.2	AIE
Minor				
201	Ø .170 +.000/-.035	Level II	3.2	AIE
202	R .030 ± .005	Level II	3.2	AIE
203	.018 min	Level II	3.2	AIE
204	Ø .316 ± .002	Level II	3.2	AIE
205	⌖ [Ø .001(M)][A][B(M)] (for Ø .316 ± .002)	Level II	3.2	AIE
206	Ø .212 ± .001 -B-	Level II	3.2	AIE
207	.033 ± .003 -A-	Level II	3.2	AIE
208	Note 4: There shall be no splits, cracks, holes, or breaks.	Level II	3.2 & 3.10	AIE
209	Note 3: Surface texture, √ 125	Level II	3.2	AIE
210	Note 3: Remove all burrs and break sharp edges; R .005 max.	Level II	3.2 & 3.10	AIE
211	Note 2: Material: plastic molding and extrusion material, cellulose acetate butyrate IAW ASTM D707, CAB, group 2, class 3 or class 4, grade 2, E11.	Level II	3.2	COC

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TABLE IV. Conformance inspection by classification of characteristics for stock and insulator assembly, drawing number 1275046

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	Note 3: The primer stock insulator, find no. 2, shall be securely held in the primer stock, find no. 1, by a 360° ring stake.	Level IV	3.2	AIE
M102	Note 4: Stock surface (datum C) shall be free of contamination and foreign material that can cause poor electrical continuity.	Level IV	3.2 & 3.10	AIE
M103	Note 6: Insulator shall not protrude aft of datum C.	Level IV	3.2	AIE
M104	Note 7: The installed insulator shall have no splits, cracks, holes, or breaks.	Level IV	3.2 & 3.10	AIE
Minor	Note 2: Surface texture, √ 125			
201	Note 2: Remove all burrs	Level II	3.2	AIE
202	Note 2: Corners at I.D. of insulator 45° x .005 max	Level II	3.2 & 3.10	AIE
203	⌖ [Ø .002(M)][A(M)] (for Ø .176 +.003/-.000)	Level II	3.2	AIE
204	Note 5: Insulator shall not protrude forward of datum B of the stock. Surface	Level II	3.2	AIE
205	texture, √ 63	Level II	3.2	AIE
206	Ø .579 -A-	Level II	3.2	AIE
207		Level II	3.2	AIE

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TABLE V. Conformance inspection by classification of characteristics for ignition element cup, drawing number 584102

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
+M101	Ø .524 ± .004	Level IV	3.2	AIE
+M102	Ø .235 ± .003 thru	Level IV	3.2	AIE
-M103	.700 +.010/-.000	Level IV	3.2	AIE
Minor				
201	Note 2: Surface texture, √ 125	Level II	3.2	AIE
202	Note 3: Material; Copper, IAW ASTM B152, composition optional, temper 060 or OS025. Alloys C12500, C14200 and C14420 shall not be used	Level II	3.2	COC
	Note 2: Remove all burrs	Level II	3.2 & 3.10	AIE
203	Ø .569 ± .001	Level II	3.2	AIE
204	Ø .538 +.002/-.000	Level II	3.2	AIE
205	.014 min	Level II	3.2	AIE
206	R .020 +.005/-.000	Level II	3.2	AIE
207	Ø .235 -.003 (Lower Tol)	Level II	3.2	AIE
208	.018 +.000/-.003	Level II	3.2	AIE
209	.156 ± .005	Level II	3.2	AIE
210	R .125 ± .005	Level II	3.2	AIE
211	Ø .524 -.004 (Lower Tol)	Level II	3.2	AIE
212	Note 2: There shall be no breaks, cracks, or splits	Level II	3.2 & 3.10	AIE
213	.700 +.010 (Upper Tol)	Level II	3.2	AIE
214	⌖ [Ø .005(M)][A(M)] (for dim Ø .235 ± .003)	Level II	3.2	AIE
215	⌖ [Ø .001(M)][A(M)] (for dim Ø .524 ± .004)	Level II	3.2	AIE
216				

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TABLE VI. Conformance inspection by classification of characteristics for ignition cup, drawing number 584098

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
+M101	Ø .300 ± .005	Level IV	3.2	AIE
M102	R .008 max	Level IV	3.2	AIE
-M103	Ø .538 +.000/-.002	Level IV	3.2	AIE
M104	R .005 max	Level IV	3.2	AIE
M105	⌖ [Ø .003(M)][A(M)]	Level IV	3.2	AIE
M106	⌖ [Ø .003(M)][A(M)]	Level IV	3.2	AIE
Minor				
201	Note 2: Surface texture, √ 125	Level II	3.2	AIE
202	Ø .300 -.005 (Lower Tol)	Level II	3.2	AIE
203	R .030 +.010/-.000	Level II	3.2	AIE
204	R .015 ± .010	Level II	3.2	AIE
205	.383 ± .001	Level II	3.2	AIE
206	.333 ± .001	Level II	3.2	AIE
207	Ø .4100 ± .0005	Level II	3.2	AIE
208	Note 3: Material, Brass IAW ASTM B36, composition optional, temper H02 or ASTM B16, temper H02.	Level II	3.2	COC
209	Note 2: Remove all burrs	Level II	3.2 & 3.10	AIE
210	Note 2: Outside corners .01 x .01 cham max	Level II	3.2	AIE
211	Ø .538 +.000 (Upper Tol)	Level II	3.2	AIE

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TABLE VII. Conformance inspection by classification of characteristics for ignition cup insulator, drawing number 584106

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
-M101	Ø .392 +.000/-.003	Level IV	3.2	AIE
Minor				
201	Note 2: Surface texture, √ 125	Level II	3.2	AIE
202	Note 3: Material; Cirlex ®, .010 inch or nylon per ASTM-D4066.	Level II	3.2	COC
203	.010 ± .002	Level II	3.2	AIE
204	Ø .103 +.003/-.000 thru	Level II	3.2	AIE
205	⌖ [Ø .002(M)][A(M)] (for dim Ø .103 +.003/-.000 thru)	Level II	3.2	AIE
206	Note 2: Remove all burrs	Level II	3.2 & 3.10	AIE
207	Ø .392 +.000 (Upper Tol)	Level II	3.2	AIE

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TABLE VIII. Conformance inspection by classification of characteristics for electrode and insulator assembly, drawing number 584087

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	Ø .174 +.000/-.002	Level IV	3.2	AIE
M102	Note 5: There shall be no breaks, cracks, holes, or other imperfections.	Level IV	3.2 & 3.10	AIE
M103	.319 ± .001	Level IV	3.2	AIE
Minor				
201	Note 3: Material: polyamide molding and extrusion material, ASTM D6779, group 5, class 1, grade 1, table S2.1, type III, grade E, A-24/23/50.	Level II	3.2	COC
202	Note 4: Material: brass, IAW ASTM B36, composition optional, temper H02 or ASTM B16, temper H02.	Level II	3.2	COC
203	.033 max .030 min	Level II	3.2	AIE
	⌖ [Ø .002(M)][B(M)] (for Ø .412 ± .001)			
204	.054 ± .002	Level II	3.2	AIE
205	Ø .412 ± .001	Level II	3.2	AIE
206	Note 5: Surface texture √ 63 all machined surfaces	Level II	3.2	AIE
207	Ø .100 ± .002	Level II	3.2	AIE
208	⊥ [Ø .001(M)][A] -B- (for Ø .174 +.000/-.002)	Level II	3.2	AIE
209	⌖ [Ø .002(M)][C(M)] (for Ø .100 ± .002)	Level II	3.2	AIE
210	.259 ± .002	Level II	3.2	AIE
211	Note 5: Outside corners .01 x .01 chamfer max	Level II	3.2	AIE
212	Note 5: Fillets R .005 max	Level II	3.2	AIE
213	Note 5: Remove all burrs	Level II	3.2	AIE
214		Level II	3.2 & 3.10	AIE

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TABLE IX. Conformance inspection by classification of characteristics for epoxy, semi-flexible, drawing number 2596952

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major M101	Note 3: Only the item described in this drawing when procured from the vendor(s) listed hereon is approved by Indian Head Division, Naval Surface Warfare Center, Indian Head, Maryland 20640-5035 for use in the application(s) specified hereon. A substitute item shall not be used without prior testing by Indian Head Division, Naval Surface Warfare Center, Indian Head, Maryland 20640-5035 or Naval Sea Systems Command.	Level IV	3.2	COC
Minor 201	Note 2: Material: this resin shall be a two part system having a mixing ratio by weight of one to one, and a specific gravity of 1.08. It shall have an initial viscosity of 7000 cps and a gel time of approximately 30 minutes at 60°C. Curing time will vary from 24 to 48 ambient temperatures to a minimum of one hour at 95°C.	Level II	3.2	COC

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TABLE X. Conformance inspection by classification of characteristics for dye, red concentrate, drawing number 3028673

<table>
<tr><th>Classification</th><th>Examination or Test</th><th>Conformance Criteria</th><th>Requirement Paragraph</th><th>Inspection Method Reference</th></tr>
<tr><td>Critical

Major

Minor
201</td><td>None Defined

None Defined

Material:

<table>
<tr><td colspan="3">Approved Source(s) of Supply</td></tr>
<tr><td>Vendor</td><td>Vendor’s Item Ident. No.</td><td>Application</td></tr>
<tr><td rowspan="3">CONAP Incorporated
Allegany, New York, 14706
Code Ident. - 16245</td><td rowspan="3">CONAP Color Concentrate 1831A</td><td>MK 1 MOD 1 Ignition Element</td></tr>
<tr><td>MK 2 MOD 0 Ignition Element</td></tr>
<tr><td>MK 15 MOD 2 Primer</td></tr>
</table>
</td><td>Level II</td><td>3.2</td><td>COC</td></tr>
</table>

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TABLE XI. Conformance inspection by classification of characteristics for ignition cup and electrode assembly, drawing number 2434751

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical				
C1	Note 6: There shall be a continuous fillet of the colored epoxy to the interface of the ignition cup insulator and the inner lip of the ignition cup as shown. The fillet shall be fully cured (rigid) and not tacky. It shall be continuous with no holes, cracks or breaks.	100%	3.2 & 3.10	AIE
Major				
M101	Note 3: A five pound min dead weight force applied to the (.100) electrode diameter [shall not unseat the electrode and electrode insulator, find no. 3, from the ignition cup and the ignition cup insulator].	Level IV	3.2	AIE
M102	Note 4: The electrode insulator shall not protrude forward of -B-.	Level IV	3.2	AIE
M103	Note 5: The assembly shall be thoroughly clean and dry prior to application of epoxy.	Level IV	3.2 & 3.10	AIE
M104	Note 7: After curing, the volume of cavity shall be a minimum of 34 microliters.	Level IV	3.2	AIE or NSIG
M105	Note 8: No epoxy shall be present on other external surfaces.	Level IV	3.2 & 3.10	AIE
M106	Note 10: The insulation resistance between the electrode and ignition cup shall be 50 megaohms min with 500 VDC min.	Level IV	3.2, 3.5 & 3.6	AIE

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TABLE XI. Conformance inspection by classification of characteristics for ignition cup and electrode assembly, drawing number 2434751 Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Major				
M107	.004 ± .004	Level IV	3.2	AIE
Minor				
201	Note 9: Minimum depth of groove on electrode shall be .005.	Level II	3.2	AIE
202	.017 +.008/-.000	Level II	3.2	AIE
203	.020 ± .005	Level II	3.2	AIE
204	⌖ [.002(M)][A(M)] (for dim .020 ± .005)	Level II	3.2	AIE
205	Ø .100	Level II	3.2	AIE
206	Ø .536 -A-	Level II	3.2	AIE
207	Note 2: The electrode and insulator assembly, find no. 3, shall be seated with the ignition cup, find no. 1, and ignition cup insulator, find no. 2.	Level II	3.2	AIE

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TABLE XII. Conformance inspection by classification of characteristics for bridge wire assembly, drawing number 2434752

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical C1	Verify cured epoxy fillet is at interface of ignition cup insulator and inner lip of ignition cup 360° around with no holes, cracks or breaks.	100%	3.2 & 3.3	AIE
Major M101	Note 3: Bridge-wire shall not touch ignition cup insulator, DWG 584106, of the ignition cup and electrode assembly, find no. 1, and shall not extend forward of datum A.	Level IV	3.2	AIE
M102	Note 4: The soldered end of the electrode shall not protrude forward of datum A of the ignition cup by more than .002.	Level IV	3.2	AIE
M103	Note 5: Bridge-wire shall not be exposed at any point on the soldered surfaces.	100%	3.2 & 3.3	AIE
M104	Note 6: The installed bridge-wire shall produce a visible red glow in both legs, simultaneously, when flashed by application of sufficient electric current.	100%	3.2	AIE
M105	Note 7: The assembly shall, after flashing, have a resistance of .10 to .18 ohms.	100%	3.2 & 3.6	AIE
M106	Note 8: The bridge-wire shall not be abraded, broken, pinched, kinked or burned after the resistance test.	100%	3.2 & 3.3	AIE
M107	Note 12A: The bridge-wire shall have a diameter of 0.0020 ± 0.0005 inches.	Level IV	3.2	AIE

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TABLE XII. Conformance inspection by classification of characteristics for bridge wire assembly, drawing number 2434752 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
M108	Note 11: Distance from outer surface of ignition cup insulator to the outer edge of the assembly (datum A), see detail A, shall be .047 minimum.	Level IV	3.2	AIE
M109	Note 12B: The wire shall have an electrical resistance ranging from 28 to 31 ohms/foot at 20°C.	Level IV	3.2, 3.5 & 3.6	COC
M110	Note 12C: Material: The wire shall be an alloy of 90% platinum, and 10% iridium, in accordance with ASTM B684 paragraph 5 (chemical requirements).	Level IV	3.2	COC
M111	The wire shall be capable of withstanding a .375 lb (6 oz) tension test without breaking when tested per ASTM F219 paragraph 9.1 (tension test apparatus) with a 12 inch gage length at a traverse rate of 12 inches per minute.	Level IV	3.2	COC
Minor				
201	Note 13: Solder, find no. 3, and flux, find no. 4, shall be IAW IPC J-STD-006 and IPC J-STD-004. Solder form is optional, impurity level B or C. Solder shall be one of the following alloys: Sn60Pb40, Sn62Pb36Ag2, or Sn63Pb37.	Level II	3.2	COC
202	Note 2: Solder bridge-wire, find no. 2, to the ignition cup and electrode assy, find no. 1, IAW IPC J-STD-001. Solder in three places as shown in main view note 2, using solder and flux, find 3 and 4. Residual flux must be washed off.	Level II	3.2	AIE
203	Note 5: There shall be no solder flux or extraneous materials on the completed assembly.	Level II	3.2	AIE

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TABLE XII. Conformance inspection by classification of characteristics for bridge wire assembly, drawing number 2434752 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Minor				
204	Note 12F: The wire shall be clean, smooth, and free from cracks or other defects that may affect its serviceability.	Level II	3.2 & 3.10	AIE
205	Note 3: Wire shall be soldered in a manner such that the wire is not taut between each solder joint.	Level II	3.2	AIE
206	Note 4: Ignition cup soldered surfaces shall be flush with their respective component surfaces after excess solder is removed.	Level II	3.2	AIE
207	Note 5: There shall not be cold solder joints.	Level II	3.2 & 3.3	AIE
208	Note 5: The solder shall adhere firmly and smoothly to the parts joined.	Level II	3.2 & 3.3	AIE
209	Note 5: The joint shall not be chalky in appearance, or lacking metallic luster.	Level II	3.2 & 3.3	AIE
210	Note 5: The joint shall not have a rough, gritty or piled-up surface.	Level II	3.2 & 3.3	AIE
211	Note 5: There shall be no solder in the ignition cavity.	Level II	3.2 & 3.3	AIE

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TABLE XIII. Conformance inspection by classification of characteristics for ignition charge, drawing number 2434757

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major M101	Note 2: Ignition charge composition: A) Normal lead styphnate……..98.00± .05% by weight B) Nitrocellulose……...2.00± .05% by weight C) Butyl acetate…….as required	Level IV	3.2	COC[1/]
M102	Note 3: The ignition charge shall be a homogeneous compound. Butyl acetate, find no. 3, shall be added as a wetting agent to make a uniformly wetted paste which is suitable for loading. No free liquid shall be present after mixing. Paste consistency shall be adjusted by addition or evaporation of the butyl acetate. Any adjustment shall be followed by a thorough remixing.	Level IV	3.2	COA
Minor	None Defined			

[1/] COC for the Ignition Compound shall include the following information for each ingredient: Specification, Manufacturer, and Manufacturer's lot number, and whether there are any deviations from the requirements of their respective specifications.

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TABLE XIV. Conformance inspection by classification of characteristics for igniter assembly, drawing number 2434754

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical C1	Note 4: No explosive shall be present on any surface of the assembly other than in the cavity.	100%	3.2 & 3.10	AIE
Major M101	Note 2: Ignition charge, find no. 2, shall be loaded until charge is flush with top of cavity and bridge-wire, find no. 1, is completely covered.	Level IV	3.2	AIE
M102	Note 3: The dried ignition charge shall be free from cracks and voids, and must not separate from cavity. The charge shall entirely cover the bridge-wire.	Level IV	3.2 & 3.10	AIE
Minor 201	Note 5: Assembly shall be maintained at 15°C min, and 60% max relative humidity until sealed into assembly, drawing 2434753.	Level II	3.2	AIE

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TABLE XV. Conformance inspection by classification of characteristics for electrode washer, drawing number 584107

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	There shall be no splits, cracks, or breaks.	Level IV	3.2 & 3.10	AIE
-M102	Ø .397 +.000/-.003	Level IV	3.2	AIE
M103	Ø .176 +.003/-.000 thru	Level IV	3.2	AIE
Minor				
201	.010 ± .002	Level II	3.2	AIE
202	Note 3: Material; nylon per ASTM D4066, or Cirlex ® .010 inch	Level II	3.2	COC
203	⌖ [Ø .002(M)][A(M)] (for dim Ø .176 +.003/-.000 thru)	Level II	3.2	AIE
204	Note 2: Remove all burrs	Level II	3.2 & 3.10	AIE
205	Note 2: Surface texture, √ 125	Level II	3.2	AIE
206	Ø .397 +.000 (Upper Tol)	Level II	3.2	AIE

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TABLE XVI. Conformance inspection by classification of characteristics for electrode washer and igniter assembly, drawing number 3208374

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major M101	Note 2: Electrode washer shall not extend over (Ø .4100) ignition cup.	100%	3.2	AIE
M102	Note 3: Electrode washer shall not be split, cracked, or broken after assembly.	Level IV	3.2 & 3.10	AIE
-M103	.006 ± .006	Level IV	3.2	AIE
Minor 201	.006 + .006 (Upper Tol)	Level II	3.2	AIE

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TABLE XVII. Conformance inspection by classification of characteristics for ignition element disc, drawing number 2846967

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	There shall be no holes, cuts, or tears.	Level IV	3.2 & 3.3	AIE
Minor				
201	Note 2: Commercial stock finish.	Level II	3.2	COC
202	Ø .530± .008	Level II	3.2	AIE
203	.020± .002	Level II	3.2	AIE
204	Material: Blotting Paper, A-A-2192, white is acceptable	Level II	3.2	COC

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TABLE XVIII. Conformance inspection by classification of characteristics for closing disc, drawing number 2846969

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	.0050± .0004	Level IV	3.2	AIE
-M102	Ø .536± .001	Level IV	3.2	AIE
M103	Note 4: There shall be no holes, splits, cracks, or breaks.	Level IV	3.2 & 3.3	AIE
Minor				
201	Note 2: Finish as rolled or punched.	Level II	3.2	AIE
202	Material: Brass, IAW ASTM B36, composition optional, temper H01 or H02	Level II	3.2	COC
203	Note 3: Remove all burrs.	Level II	3.2 & 3.10	AIE

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TABLE XIX. Conformance inspection by classification of characteristics for spacer, drawing number 584099

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
+M101	R .035 ± .005	Level IV	3.2	AIE
-M102	Ø .536 ± .001 -A-	Level IV	3.2	AIE
Minor				
201	⌖ [Ø .005(M)][A(M)] (for Ø .300 ± .005)	Level II	3.2	AIE
202	Ø .300 ± .005	Level II	3.2	AIE
203	.210 ± .001	Level II	3.2	AIE
204	45° x .010 ± .005	Level II	3.2	AIE
205	R .035 -.005 (Lower Tol)	Level II	3.2	AIE
206	Ø .536 +.001 (Upper Tol)	Level II	3.2	AIE
207	Note 2: Surface texture, √ 125	Level II	3.2	AIE
208	Note 2: Remove all burrs	Level II	3.2 & 3.10	AIE
209	Material: Brass, IAW ASTM B36, Composition optional, Temper H02, or ASTM B16, Temper H02	Level II	3.2	COC

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TABLE XX. Conformance inspection by classification of characteristics for ignition element, electric; MK 1 MOD 1 assembly, drawing number 2434753

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical				
C1	Note 8: There shall be no explosives on exterior surfaces.	100%	3.2 & 3.10	AIE
Major				
M101	Note 3: The electrode washer and igniter assembly, shall be dried 4 hours minimum at a temperature of 38° to 49°C immediately preceding assembly.	Level IV	3.2	AIE
M102	Note 4: Class 4 black powder, find no. 6, charge shall be 3.3 grains min. Moisture content of black powder shall be 0.70% max.	Level IV	3.2 & 3.4	AIE
M103	Note 6: Assembly shall have 360° crimp.	Level IV	3.2	AIE
M104	Note 7: There shall be no holes, cracks, splits, or breaks in the ignition element cup or exposed closing disc, find no. 4, after assembly. The closing disc shall be under the crimped ignition element cup 360°.	100%	3.2 & 3.3	AIE
M105	Note 9: The assembly shall have an electrical resistance of .10 to .18 ohms.	100%	3.2, 3.5 & 3.6	AIE

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TABLE XX. Conformance inspection by classification of characteristics for ignition element, electric; MK 1 MOD 1 assembly, drawing number 2434753 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Major				
M106	Note 10: Immediately prior to next assembly apply moisture proof varnish, find no. 7, 360° around in two places as shown for seal.	Level IV	3.2	AIE
M107	.231 +.007/-.004	Level IV	3.2	AIE
M108	⌖ [Ø .009(M)][A(M)] (for Ø .569 +.001/-.003)	Level IV	3.2	NSIG
M109	Ø .569 +.001/-.003	Level IV	3.2	AIE
M110	.025 min	Level IV	3.2	AIE
M111	.652 ± .016	Level IV	3.2	AIE
M112	R .020 min	Level IV	3.2	AIE
M113	R .010 min	Level IV	3.2	AIE
M114	Note 5: Component parts shall be clean and dry prior to assembly to avoid accumulation of moisture within the ignition element.	Level IV	3.2 & 3.10	AIE
Minor				
201	Ø .174 -A-	Level II	3.2	AIE
202	Note 2: Ignition element cup, find no. 1, shall be assembled to electrode washer and igniter assembly, find no. 2.	Level II	3.2	AIE

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TABLE XXI. Conformance inspection by classification of characteristics for primer tube, drawing number 685475 and MIL-DTL-15119A

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Drawing 685475				
Critical				
C1	Note 3: Hydrostatic test to 18,000 psi min, 19,000 psi max			
-C2	.750-32UN-2A -A-	100%	3.2	AIE
C3	32 holes	100%	3.2	AIE
C4	Ø .5990 +.0000	100%	3.2	AIE
	Also, see C1 in MIL-DTL-15119, the specification for steel tubing	100%	3.2	AIE
Major				
M101	Ø .218 ± .010			
M102	⌖[Ø .020(M)][C(M)][B]	Level IV	3.2	AIE
M103	Ø .580 +.000/-.010	Level IV	3.2	AIE
M104	.830 max runout	Level IV	3.2	AIE
		Level IV	3.2	AIE
M105	Note 4: Zinc plate IAW ASTM B633, SC 1, type II, maximum thickness of 20µm. Threads are not subject to minimum thickness requirement, but shall show evidence of coating.	Level IV	3.2	COA (for thickness)/AIE
M106	Ø .760 +.010/-.000			
M107	.740 min full thread	Level IV	3.2	AIE
M108	8.250 ± .020	Level IV	3.2	AIE
-M109	Ø .5990 -.0007	Level IV	3.2	AIE
+M110	.750-32UN-2A	Level IV	3.2	AIE
M111	⌖ [Ø .003(S)][A(M)]	Level IV	3.2	AIE
		Level IV	3.2	AIE
Minor				
201	Note 2: Finish as drawn, surface texture √ 125 for machined surfaces			
		Level II	3.2	COC
202	Note 3: Material, see MIL-DTL-15119 requirements; contains critical and major characteristics	Level II	3.2	AIE

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TABLE XXI. Conformance inspection by classification of characteristics for primer tube, drawing number 685475 and MIL-DTL-15119A Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Drawing 685475 (continued)				
Minor				
203	R .04			
204	.085 min	Level II	3.2	AIE
205	.240 ± .010	Level II	3.2	AIE
206	.055 min	Level II	3.2	AIE
207	Ø .630 ± .008	Level II	3.2	AIE
208	— [Ø .030 (M)] (for Ø .760 +.010/-.000)	Level II	3.2	AIE
209	.05 ± .02	Level II	3.2	AIE
210	20.250 ± .020	Level II	3.2	AIE
211	R .025 or 45° x .025	Level II	3.2	AIE
212	1.00 ± .03	Level II	3.2	AIE
213	45° x .020 ± .005	Level II	3.2	AIE
214	Note 2: Fillets R .010 max	Level II	3.2	AIE
215	Note 2: Outside corners 45° x .02 max	Level II	3.2	AIE
216	Note 2: Remove all burrs	Level II	3.2	AIE
217	15 x .75	Level II	3.2 & 3.10	AIE
		Level II	3.2	AIE
MIL-DTL-15119				
Critical				
C1	Hydrostatic test to 18,000 PSI min.	100%	3.2	AIE
Major				
M101	Manufacture. The tubing shall be drawn from fully killed steel manufactured by either the open hearth or electric furnace process.	Level IV	3.2	COC

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TABLE XXI. Conformance inspection by classification of characteristics for primer tube, drawing number 685475 and MIL-DTL-15119A Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
MIL-DTL-15119 (continued)				
Major				
M102	Chemical composition (4130 steel)	Level IV	3.2	COA
M103	Grain size	Level IV	3.2	COA
M104	Heat Treatment / Mechanical Properties	Level IV	3.2	COC
M105	Dimensions (per contract)	Level IV	3.2	AIE
M106	Tolerances (per contract)	Level IV	3.2	AIE

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TABLE XXII. Conformance inspection by classification of characteristics for primer tube liner, drawing number 2847020

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	.0035 +.0010/-.0005	Level IV	3.2	AIE
+M102	2.500 ± .030	Level IV	3.2	AIE
Minor				
201	18.00 ± .06	Level II	3.2	AIE
202	Note 2: Material; bond paper, A-A-2804, type II. Approximate weight of 1000 sheets (17 x 22) is 32 lbs. Alternate: paper, sealing IAW MIL-P-60169, type II.	Level II	3.2	COC
203	Grain of paper ↕	Level II	3.2	AIE
204	2.500 -.030 (Lower Tol)	Level II	3.2	AIE

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TABLE XXIII. Conformance inspection by classification of characteristics for plug, drawing number 685405

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
+M101	Ø .700 +.000/-.004	Level IV	3.2	AIE
-M102	1.217	Level IV	3.2	AIE
M103	.302 ± .005	Level IV	3.2	AIE
M104	.125 +.002/-.000	Level IV	3.2	AIE
+M105	Chamfer 45° x .030 ± .010	Level IV	3.2	AIE
M106	Ø .584 +.003/-.000	Level IV	3.2	AIE
M107	Ø .200 thru	Level IV	3.2	AIE
M108	R .02 max	Level IV	3.2	AIE
M109	Ø .5993 +.0005/-.0000	Level IV	3.2	AIE
Minor				
201	Note 2: Surface texture, √ 125	Level II	3.2	AIE
202	⌖ [Ø .002 (M)][A(S)] (for dim Ø .700 +.000/-.004)	Level II	3.2	AIE
203	⌖ [Ø .003 (M)][A(S)] (for dim Ø .584 +.003/-.000)	Level II	3.2	AIE
204	R .015	Level II	3.2	AIE
205	⌖ [Ø .010 (M)][A(M)] (for dim Ø .200 thru)	Level II	3.2	AIE
206	45° x .05	Level II	3.2	AIE
207	45° x .005 max	Level II	3.2	AIE
208	R .01 max	Level II	3.2	AIE
209	30°	Level II	3.2	AIE
210	.125 +.010/-.000	Level II	3.2	AIE
211	⊥ [.001][A]	Level II	3.2	AIE
212	Note 3: Material; Brass, IAW ASTM B36, composition optional, temper H02 to H04, or ASTM B16, temper H02 to H04.	Level II	3.2	COC

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TABLE XXIII. Conformance inspection by classification of characteristics for plug, drawing number 685405 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Minor				
213	Note 2: Remove all burrs	Level II	3.2 & 3.10	AIE
214	Note 2: Outside corners 45° x .01 max	Level II	3.2	AIE
215	R .015	Level II	3.2	AIE
216	1.217 (+M102)	Level II	3.2	AIE
217	.700 +.000/-.004 (-M101)	Level II	3.2	AIE
218	.030 (-M105)	Level II	3.2	AIE
219	45°	Level II	3.2	AIE

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TABLE XXIV. Conformance inspection by classification of characteristics for primer tube, liner and plug assembly, drawing number 685477

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
M101	Note 3: No paper liner shall be visible in holes, see notes 8, 9, and 10. Pin holes and cracks in the sealant which can be penetrated by a pin of Ø .020 inch diameter to a depth of .050 inch are not acceptable.	Level IV	3.2	AIE
M102	Note 4: Plug, find no. 4, shall be press fit into tube, and shall be seated against tube at, at least one point.	Level IV	3.2	AIE
M103	Note 5: After assembly, the liner shall not be torn or crumpled.	Level IV	3.2	AIE or NSIG
M104	Major Ø .7499 max	Level IV	3.2	AIE
M105	Note 6: The (Ø .200) hole in the plug after assembly with all other parts shall be [free of all obstructions].	Level IV	3.2	AIE
M106	⌖ [Ø .005(M)][A(M)] (for dim Ø .584)	Level IV	3.2	AIE
Minor				
201	.15 max	Level II	3.2	AIE
202	Purple Lacquer IAW MIL-L-296	Level II	3.2	COA
203	After inserting liner, holes shall be sealed with purple lacquer.	Level II	3.2	AIE

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TABLE XXV. Conformance inspection by classification of characteristics for closing cup, drawing number 2847019

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
-M101	.016± .004	Level IV	3.2	AIE
M102	"Constrained state" requirement: when assembled in the primer tube liner and plug assembly, .240 groove, the side walls of the cup shall form a 360° seal without forming any furrows, gaps or open channels. Navy gage no. 6656381 may be used to verify the constrained state.	Level IV	3.2	NSIG
Minor				
201	R .080 +.030/-.040	Level II	3.2	AIE
202	◯ [.01] (F) (for dim .080 +.030/-.040)	Level II	3.2	AIE
203	.230 ± .020	Level II	3.2	AIE
204	Ø .630 +.020/-.000 AVG	Level II	3.2	AIE
205	.600 +.005/-.015 (T) to R .080	Level II	3.2	AIE
206	◯ [.01] (F) (for dim .600 +.005/-.015 (T) to R .080)	Level II	3.2	AIE
207	.030 +.014/-.014 AVG	Level II	3.2	AIE
208	.016 +.004 (Upper Tol)	Level II	3.2	AIE
209	Note 2: Material: insulation, electrical, paper, MIL-I-695, Type R.	Level II	3.2	COC
210	Note 3: Dimensions apply to closing cup in free state at the time it is manufactured.	Level II	3.2	AIE

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TABLE XXV. Conformance inspection by classification of characteristics for closing cup, drawing number 2847019 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Minor 211	Note 3: Over time, cup dimensions may shift due to release of internal stresses and effects of humidity. However, the cup will still satisfy its functional requirements at the next level of assembly, which is verified by satisfying the "constrained state" (see note 5).	Level II	3.2	NSIG
212	Note 4: The side walls of the cup shall be pliable to allow proper function of the cup in the next higher assembly, as described in the "constrained state" requirement.	Level II	3.2	NSIG
213	Note 6A: Verification requirements of MIL-DTL-18714 shall be met by certification of compliance for material; note 2.	Level II	3.2	COC
214	Note 6B: Verification requirements of MIL-DTL-18714 shall be met by manufacturer's data showing dimensional compliance in "free state"; note 3.	Level II	3.2	AIE
215	Note 6C: Verification requirements of MIL-DTL-18714 shall be met by compliance of "constrained state"; note 5.	Level II	3.2	NSIG
NOTE	Verification requirements for first article inspection shall be met by the "constrained state". Manufacturer's data showing dimensional compliance at time of manufacture pertains to the "free state".	Level IV & II	3.2	AIE/NSIG

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TABLE XXVI. Conformance inspection by classification of characteristics for cup, inner, drawing number 3208399

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None Defined			
Major				
-M101	.016 ± .004	Level IV	3.2	AIE
M102	Ø .610 +.020/-.000 AVG	Level IV	3.2	AIE
M103	"Constrained state" requirement: when assembled in the mating primer tube liner and plug assembly, the side walls of the cup shall form a 360° seal without forming any furrows, gaps or open channels. Navy gage no. 6656381 may be used to verify the constrained state.	Level IV	3.2	NSIG
Minor				
201	.510 +.030/-.010 (T) to R .080	Level II	3.2	AIE
202	◯ [.01] (F) (for dim .510 +.030/-.010 (T) to R .080)	Level II	3.2	AIE
203	R .080 +.040/-.030	Level II	3.2	AIE
204	.016 + .004 (Upper Tol)	Level II	3.2	AIE
205	.500 ± .030	Level II	3.2	AIE
206	.045 +.015/-.030 AVG	Level II	3.2	AIE
207	Note 2: Material: insulation, electrical, paper, MIL-I-695, Type R.	Level II	3.2	COC
208	Note 3: Dimensions apply to cup, inner in "free state" at the time it is manufactured.	Level II	3.2	AIE

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TABLE XXVI. Conformance inspection by classification of characteristics for cup, inner, drawing number 3208399 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Minor 209	Note 3: Over time, cup dimensions may shift due to release of internal stresses and effects of humidity. However, the cup will still satisfy its functional requirements at the next level of assembly, which is verified by satisfying the "constrained state" (see note 5).	Level II	3.2	NSIG
210	Note 4: The sidewalls of the cup shall be pliable to allow proper function of the cup in the next higher assembly, as described in the "constrained state" requirement.	Level II	3.2	NSIG
NOTE	Verification requirements for first article inspection shall be met by the "constrained state". Manufacturer's data showing dimensional compliance at time of manufacture pertains to the "free state".	Level IV & II	3.2	AIE/NSIG

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TABLE XXVII. Conformance inspection by classification of characteristics for primer electric, MK 45 MOD 1 assembly, drawing number 2434755

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical				
C1	Note 3: Prior to assembly:			
	(A) Mating surfaces shall be free of contamination or foreign material.	100%	3.2 & 3.10	AIE
	(B) Surface primer, find no. 4, shall have been applied to mating threads of stock and insulator assembly, find no. 1, and primer tube, liner and plug assembly, find no. 3, and dried for 15 minutes minimum at 70 ± 10°F minimum.	100%	3.2	AIE
	(C) Primer tube threads shall be coated 360° around with sealing compound, find no. 5. Sufficient sealing compound shall be applied so that a minimum of 75% of the engaged thread will be coated.	100%	3.2	AIE
C2	Note 4B: The stock and insulator assembly with ignition element installed shall not rotate on the primer tube, liner and plug assembly when 22 ± 2 ft-lbs reverse torque is applied.	100%	3.2	AIE
C3	9.1 min	100%	3.2	AIE or NSIG
C4	Note 14: No explosive particles shall be present on exterior surfaces.	100%	3.2 & 3.10	AIE

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TABLE XXVII. Conformance inspection by classification of characteristics for primer electric, MK 45 MOD 1 assembly, drawing number 2434755 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Major				
M103	Note 7: There shall be no sealing compound on primer electrode or the external threads of the primer stock.	Level IV	3.2	AIE
M104	Note 5: Electrode and insulator shall not protrude forward of stock surface, datum A. 100% inspection required.	100%	3.2	AIE
M105	Note 6: Depth of electrode below stock surface, datum A, shall be .018 max. 100% inspection.	100%	3.2	AIE
M106	Note 8: Black powder shall be checked during primer production to assure conformance with MIL-DTL-18714, paragraph 3.4.	Level IV	3.2 & 3.4	AIE
M107	Note 9: Class 2 black powder charge shall be [50.5 – 53.1 grams].	Level IV	3.2	AIE
M108	Note 10: During assembly, inner cup, find no. 9, shall be located firmly against black powder. Cup shall not be cocked or tilted. Primer tube liner shall not be torn or crumpled.	Level IV	3.2	AIE
M109	Note 11: After application of approximately 3 cc of type B moisture proof varnish, find no. 8, to inner cup so that the lip of the cup is covered by varnish, the primer shall be inverted and permitted to drain for 15 minutes minimum. A bead of varnish shall be formed 360° around the lip of the inner cup.	Level IV	3.2	AIE

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TABLE XXVII. Conformance inspection by classification of characteristics for primer electric, MK 45 MOD 1 assembly, drawing number 2434755 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Major				
M110	Note 12: The closing cup, find no. 7, shall be completely coated with type B moisture proof varnish, find no. 8, and shall be expanded against inside wall of the groove in tube. A bead of varnish shall be formed 360° around the lip of the closing cup.	Level IV	3.2	AIE
M111	Note 13: Etch 13A and 13B information on surface, datum A. The characters shall be .06 min size to a depth that is clearly and permanently legible. Ammunition lot no. shall be in accordance with MIL-STD-1168. (A) Item nomenclature: MK 45 MOD 1 or MK 45-1. (B) Ammunition lot number.	Level IV	3.2 & 3.10	AIE
M112	Note 15: The complete assembly shall have an electrical resistance of .10 to .18 ohms. 21.363 +.000/-.089	100%	3.2, 3.5 & 3.6	AIE
M113		Level IV	3.2	AIE
M114	Note 18: There shall be no varnish contamination or any foreign material on the electrode, insulator or stock.	Level IV	3.2 & 3.10	AIE
M115	Note 4A: Stock and insulator assembly, with ignition element installed, shall be torqued to the primer tube, liner and plug assembly with 27±3 ft-lbs.	Level IV	3.2	AIE
M116	Note 21: All primers must be radiographically inspected to satisfy radiographic requirements of MIL-DTL-18714.	100%	3.2 & 3.3	AIE

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TABLE XXVII. Conformance inspection by classification of characteristics for primer electric, MK 45 MOD 1 assembly, drawing number 2434755 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Minor				
201	Note 2: Ignition element, find no. 2, shall be sealed in two places, 360° around, with moisture proof varnish, find no. 8.	Level II	3.2	AIE
202	Note 16: Varnish shall be dry prior to packaging.	Level II	3.2	AIE
203	Note 19: The manufacturer shall take precautions during loading to ensure the black powder in the loaded primer does not exceed the maximum moisture content requirement (0.70% by weight).	Level II	3.2 & 3.4	AIE
204	Surface Primer, IAW ASTM D5363, Grade N.	Level II	3.2	COC
205	Black Powder, Class 2, IAW MIL-P-223.	Level II	3.2	COA
206	Sealing Compound, IAW ASTM D5363, AN0121 or AN0123.	Level II	3.2	COC
207	Moisture proof varnish, IAW MIL-V-16399.	Level II	3.2	COC
208	Note 2: Varnish shall not prevent contact between the primer stock inner face and the outer face of the ignition element cup, electrode end (2A).	Level II	3.2	AIE
209	Note 15: The stock and the electrode shall be used as terminals. 15 to 20 lb force shall be applied to the electrode during resistance measurement.	Level II	3.2	AIE

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TABLE XXVII. Conformance inspection by classification of characteristics for primer electric, MK 45 MOD 1 assembly, drawing number 2434755 – Continued

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Minor				
210	Note 15: Test circuit which limits the bridge current to 50 milliamperes max.	Level II	3.2	AIE
211	Note 19: Primer shall be loaded in a controlled humidity environment.	Level II	3.2	AIE
212	Note 4A: Assembly shall not be subjected to excessive rough handling for four hours minimum.	Level II	3.2	AIE
213	Note 4B: After minimum of 24 hours at 70±10°F has elapsed since the sealing compound has been applied	Level II	3.2	AIE

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TABLE XXVIII. Detail strip requirements from primer, electric MK 45 MOD 1 assembly, drawing number 2434755

Classification	Examination or Test
	Note: The detail strip shall be performed by the government facility as designated in the contract or order. This table is intended for informational purposes to mitigate contractor concerns in meeting requirements and serves as a reference for what the government will be verifying later.
C2	Verify the primer stock does not rotate on the primer tube when 20 ft-lbs +/- 10%, reverse torque is applied.
C3	Verify the distance from the face of the primer stock to the first vent hole is at least 9.1 inches.
C4	Verify no explosive particles are present on exterior surfaces.
M110	Verify the closing cup has 360° bead of varnish around lip of the cup.
M111	Verify the indent stamp on face of the primer stock is plainly legible, and contains MK/MOD, and manufacturer's lot number.
M103	Verify no sealing compound on primer electrode or the external threads of the primer stock.
M104	Verify electrode and insulator does not protrude forward of stock surface.
M105	Verify depth of electrode is no greater than .018", below primer stock surface.
M112	Verify complete assembly for electrical resistance of .10 to .18 ohms when measured by a test circuit limiting bridge current to 50 milliamps max. Use primer stock and electrode as terminals. Apply 15 to 20 lbs-force to electrodes during resistance measurement.
M113	Verify the assembled length to be 21.363" +.000/-.089.
M114	Verify no varnish contamination or any foreign material on the electrode, insulator or stock.

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TABLE XXVIII. Detail strip requirements from primer, electric MK 45 MOD 1 assembly, drawing number 2434755 – Continued

Classification	Examination or Test
	Post disassembly of the stock and insulator assembly (1275046) from the primer tube, liner & plug assembly (3685477), removal of the black powder, and closing cup, verify the following:
C1	Primer tube shows evidence of sealing compound on 75% of the engaged thread, 360° around.
M109	Inner cup is sealed 360° around the lip of the cup.
M106	Provide black powder sample for moisture analysis: Randomly select 5 primers, 10 grams/primer, and deliver to lab.
M107	Verify weight of class 2, main black powder charge, is 50.5 - 53.1 grams.

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4.6 Methods of inspection.

4.6.1 Radiographic. The primer shall be subjected to a radiographic inspection in accordance with ASTM E1742. The primer shall be mounted in a vertical position and radiographically inspected perpendicular to its axis, focused at a point 0.6 ± 0.2 inch from the stock face. During first article inspection or lot acceptance inspection, any primer that appears to have been misassembled shall be subjected to detail strip and shall meet the requirements of 3.5.

4.6.2 Determination of moisture content. Moisture analysis procedure for the black powder shall be as follows:

a. 10 grams of black powder from each sample primer shall be tested in the granular form for moisture content.

b. The black powder shall be removed from the primer tubes and placed in a dry container with a tight cover. Black powder shall not be exposed to the atmosphere for more than 3 minutes.

c. Sample containers must be covered and placed in a desiccator until ready to test the moisture content.

d. Carefully weigh 2.0 ± 0.1 g of sample into a tared weighing dish or container. The sample shall be weighed to the nearest 0.1 mg. Black powder shall not be exposed to the atmosphere for more than 3 minutes.

e. Uncovered samples shall then be placed in an oven at 70 to 75°C. Do not draw vacuum.

f. After 4 hours minimum (4 ½ hours maximum), remove samples from oven, cover, and place in a desiccator to cool. After cooling, carefully weigh samples to the nearest 0.1 mg. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{E}{C} \text{ x } 100$$

Where

C = A - B
E = A – D

and

A = Sample weight plus container weight, un-dried, g
B = Un-dried container weight, g
C = Un-dried sample weight, g
D = Sample weight plus container weight, after drying, g
E = Weight difference, after drying, g

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4.6.3 Detail strip. The primer shall be disassembled by the government facility designated in the contract or order to determine conformance to the requirements of Table XXVIII.

4.6.4 Electrical resistance. The primer electrical resistance shall be measured by using a test circuit which limits the bridge current to 50 milliamperes maximum. The primer stock and the electrode shall be used as terminals. 15-20 lb-force shall be applied to the electrode during resistance measurement with a probe having a 0.050 ± 0.005 inch spherical radius. Assure that contact on clean bright surfaces is obtained.

4.6.5 Function. The government facility designated in the contract or order shall evaluate the primer as described in 4.6.5.1 through 4.6.5.3.2.

4.6.5.1 Test conditions. The test for function shall be conducted at 70°F ± 20°F. The test fixture shall hold the primer securely at the primer stock external threads with no other support.

4.6.5.2 Ignition. The primer shall be initiated by the application of an electrical current through a steel firing pin with a tip spherical radius of 0.127 ± 0.013 cm (0.050 ± 0.005 inch). The firing pin tip shall be free of corrosion or foreign materials. The firing pin tip shall apply 71.172 - 97.861 Newtons (16-22 lb-force) of axial force to the primer electrode. The power supply shall provide an open circuit, direct current potential of 20.0 ± 0.5 volts dc at the primer fixture. The closing of firing key (CFK) mechanism shall be a solid-state firing switch. The firing circuit resistance shall be adjusted to provide 10 ± 0.2 amps with a current rise time of 50 microseconds maximum after CFK and shall be able to maintain the current requirement for 0.1 seconds minimum.

4.6.5.3 Data measurement. The primer function time shall be measured from the application of current (fire power) until the first indication of light at either hole of the first pair of vent holes on the primer. The light detector units (see 4.6.5.3.2) shall be placed as indicated in Figure 2.

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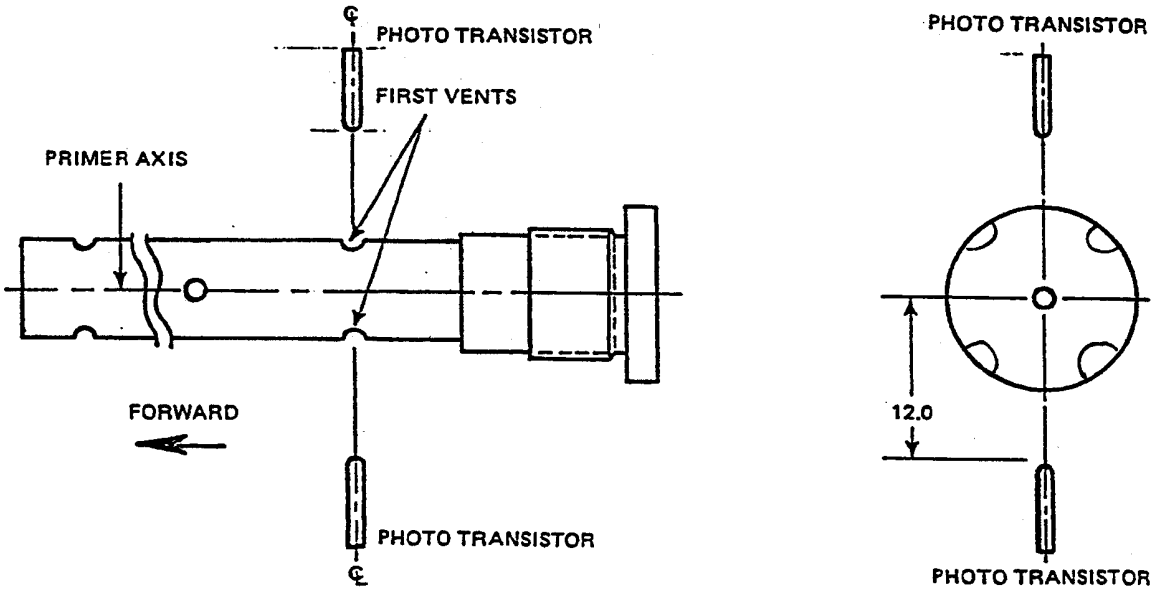


Photo Transistor Location

FIGURE 2. Light detector setup diagram

4.6.5.3.1 Data acquisition equipment. To verify the 50 microsecond rise time of the fire pulse, the data acquisition equipment shall digitize the analog voltage signals at a minimum rate of 100,000 samples per second. The data acquisition equipment shall also display or provide a plot of the fire pulse signal, so that the pulse rise time can be measured. The recommended sampling rate is at or above 10,000,000 samples per second for 100 microseconds producing 1,000 samples that completely capture the entire fire pulse rise.

The function time measurement data acquisition equipment shall digitize two channels of analog voltage signals from the light detectors at a minimum rate of 50,000 samples per second per channel for 10 milliseconds after the initial fire pulse producing at least 500 samples at an accuracy of 20 microseconds. The data acquisition equipment shall also display or provide a plot of the two light detector signals so that the function time can be measured, and store the digital data. The recommended rate is at or above 1,000,000 samples per second for 10 milliseconds producing 10,000 samples at an accuracy of 1 microsecond.

The vertical accuracy of this scope or digitizer shall be a minimum of 8 bits. A 12 bit or better unit is recommended. The sensitivity of the recording scope or digitizer should be set so that the vertical span of the data pulses should be at least half of the vertical span of the recording scope or digitizer.

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4.6.5.3.2 Light detectors. The light detectors shall respond to light of wavelength 0.7-1.0 micrometers within a range of 9 mW/cm^2. The light detector shall respond to the bright light of the primer output within 10 microseconds producing a greater than 0.5 volt swing. The light detector unit shall be capable of withstanding the shock and blast of at least 130 shots before major repairs. The recommended unit is a combination of light detector and amplifier built into a three wire IC such as a TSL 254 or TSL 254R, Precision High Speed Light to Voltage Converter by Texas Advanced Optoelectronic Solutions. It produces a 3.5-volt swing for $1mW/cm^2$ of light in the wavelength of 0.3-1.1 micrometers. A dark film or lens in front of the detector may be used to attenuate the light signal and provide a more stable data signal.

4.6.5.4 Marginality of success inspection.

4.6.5.4.1 Post firing, primers shall be visually inspected to assure compliance with the requirements of 3.7 c and 3.7 d.

4.6.5.4.2 Post firing, primers shall be disassembled and inspected to assure compliance with the requirements of 3.7 d.

4.6.6 Vibration. The primer shall be subjected to the transportation vibration test of MIL-STD-331, Test B1, section 6.1.2, table B1-4, except that the orientation of each primer during this three-phase test shall be as follows:

a. Phase I: Primer axis horizontal;

b. Phase II: Primer axis vertical, primer stock down;

c. Phase III: Primer axis vertical, primer stock up.

In order to determine the integrity of the primer after vibration, the primer shall be marked with a line in the axial direction at the junction of the primer stock and primer tube indexing their initial relative positions. The primer shall be rigidly mounted to the vibration table at the primer stock. After vibration, the primer shall be subjected to the tests of 4.6.4 and 4.6.5.

4.6.7 Ballistics. The primer shall be assembled into a 5"/54 caliber full service propelling charge assembly in accordance with drawing 3208375. All components shall be from service acceptable lots. The propellant shall be conditioned at 90 ± 15°F for 24 hours minimum prior to assembly of the propelling charge. All other components shall be conditioned at 85 ± 20°F for 24 hours minimum prior to assembly of the propelling charge. The assembled propelling charge shall be conditioned at 90 ± 5°F for 48 hours minimum. After removal from the conditioning chamber, the assembled propelling charge shall be fired within 10 minutes. Serviceable blind, loaded, and plugged projectiles shall be used. The rounds shall be loaded into a 5”/54 or 5”/62 gun using the gun’s automatic handling and ramming system, then fired using the gun’s electrical firing system. Projectile ejection time shall be recorded.

4.6.8 Inspection of packaging. The packaging, packing, and marking shall be examined for conformance with the technical data package.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory)

6.1 Intended use. The MK 45 MOD 1 Electric Primer is intended for use as the initiator in the MK 67 MODs, 5"/54 caliber full propelling charge. Since the primer was developed strictly for use in a military gun charge, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification;

b. If required, the specific issue of individual documents referenced (see 2.2 and 2.3);

c. Whether a first article sample is required (see 3.1);

d. The facility/facilities designated to perform first article inspection (see 4.3);

e. The facility/facilities designated to perform quality conformance inspection if other than the contractor (see 4.4.2);

f. Lot quantity if other than specified in 4.4.1;

g. Inspection sample quantity if other than specified in 4.4.2 and 4.4.3;

h. Test and examination procedures and equipment to be used (see 4.2 and 6.6);

i. Special marking, if required (see 6.4.4);

j. Ammunition data cards (see 6.4.5);

k. Safety precautions (see 6.5);

l. Packaging (6.4).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a pre-production sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Packaging.

6.4.1 Packaging (Level A). The primer should be packaged in Primer Pack, MK 43 MOD 0, ADL 5166053. 24 units of desiccant, conforming to Type I of MIL-D-3464, should be placed in each container prior to securing the cover.

6.4.1.1 Inert components. Components and sub-assemblies should be packaged in commercial fiberboard containers meeting the requirements of ASTM D5118. Each container should incorporate corrugated or solid fiberboard dividers or collars suitable for preventing damage in handling and transportation. The gross weight of the container contents should not be greater than 70 pounds.

6.4.2 Packing (Level A). Primers packaged in accordance with 6.4.1 should be packed in accordance with drawing 5166057.

6.4.3 Palletization. Palletization of primers packed in accordance with 6.4.2 should be in accordance with WR-53/884.

6.4.4 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), all markings should be in accordance with MIL-STD-129, and drawing 5012851, and the appropriate sections of 49 CFR.

6.4.5 Ammunition data cards. When specified in the contract or purchase order (see 6.2), 1 ammunition data card prepared in accordance with MIL-STD-1168 should be placed in the top of each pack (see 6.4.1); 5 cards should be forwarded to the contracting activity; and 20 should go to the ammunition depot receiving the primers. For Navy acquisitions, 2 data cards should also go to each of the activities listed in 6.6.

6.4.6 Ammunition lot number. Ammunition lot number should be in accordance with MIL-STD-1168.

6.5 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP-5) should be made applicable.

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6.6 Test procedures and reporting. For Navy acquisitions, the contracting activity should include in each work authorization for first article and quality conformance inspections, the requirement for the preparation and forwarding to the activities listed below of all test and examination procedures prior to performance of any inspection. The work authorization should also include the requirement for the forwarding of all test data in an intelligible form to the activities listed below within five working days of the completion of the tests.

US Navy Major Caliber Propelling Charge AEA,
Naval Surface Warfare Center Indian Head Division
Picatinny Detachment, Attn: G2, Picatinny, NJ 07806-5000.)

Department of the Navy, Naval Surface Warfare Center, Indian Head Division
Attn: Gun Propelling Charge Design Agent, Code E14,
3767 Strauss Ave, Suite 201, Indian Head MD. 20640-5150

6.7 Definitions and abbreviations.

6.7.1 Misfire. Misfire is the failure of the main black powder charge of the primer to ignite when the primer is subjected to the appropriate firing source.

6.7.2 Blowback. Blowback is gas leakage around or through the stock end of the primer and is verified by observance during firing or by visual evidence of residue after firing.

6.7.3 AIE. Acceptance Inspection Equipment.

6.7.4 COC. Certificate of Compliance.

6.7.5 COA. Certificate of Analysis.

6.7.6 NSIG. Navy Supplied Inspection Gage.

6.7.7 Homogeneous. When all units of product in an ammunition lot have been produced by one manufacturer, in one unchanged process, under stable conditions of production, in accordance with the same drawings, the same specifications and any revisions thereto, a "state of homogeneity" will be considered to exist. This means that the items of ammunition have been manufactured or assembled during a production process which has not been altered by innovation, changes in materiel sources, strikes, retooling (other than that done due to routine changes to compensate for normal tool wear or breakage) or interruptions other than those due to the end of the shift, day or week.

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6.8 Subject term (key word) listing.

Black powder
Initiator
Propelling charge

6.9 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

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